OpenMRS
Product Teardown

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AI Technical Product Manager Portfolio

“The PM who understands the legacy system will build the better AI product. Most AI-in-healthcare startups treat the EMR as a black box. OpenMRS teaches you what is inside the box.”

Executive Summary

OpenMRS is an open-source electronic medical record system designed for healthcare settings in low and middle-income countries. It emerged in 2004 out of global health and informatics collaboration, with roots in work from the Regenstrief Institute and long-standing partnership from organizations like Partners in Health.

According to OpenMRS.org, the platform has an estimated development value of $70 million and received over 2,200 code contributions in 2025. It is deployed across healthcare facilities in over 40 countries, with country-specific distributions including KenyaEMR, UgandaEMR, and eSaude (Mozambique). It is backed by a broad coalition of global health organizations and implementation partners.

This teardown fills a gap that no other product in this portfolio covers. While Cursor, Perplexity, and Claude represent the AI-native layer, OpenMRS represents the healthcare infrastructure layer that any AI product must integrate with.

Problem and Why OpenMRS Exists

In the early 2000s, the HIV/AIDS pandemic was devastating developing countries. Healthcare facilities often had no digital record systems. Patient histories were on paper. Research was nearly impossible because there was no structured data.

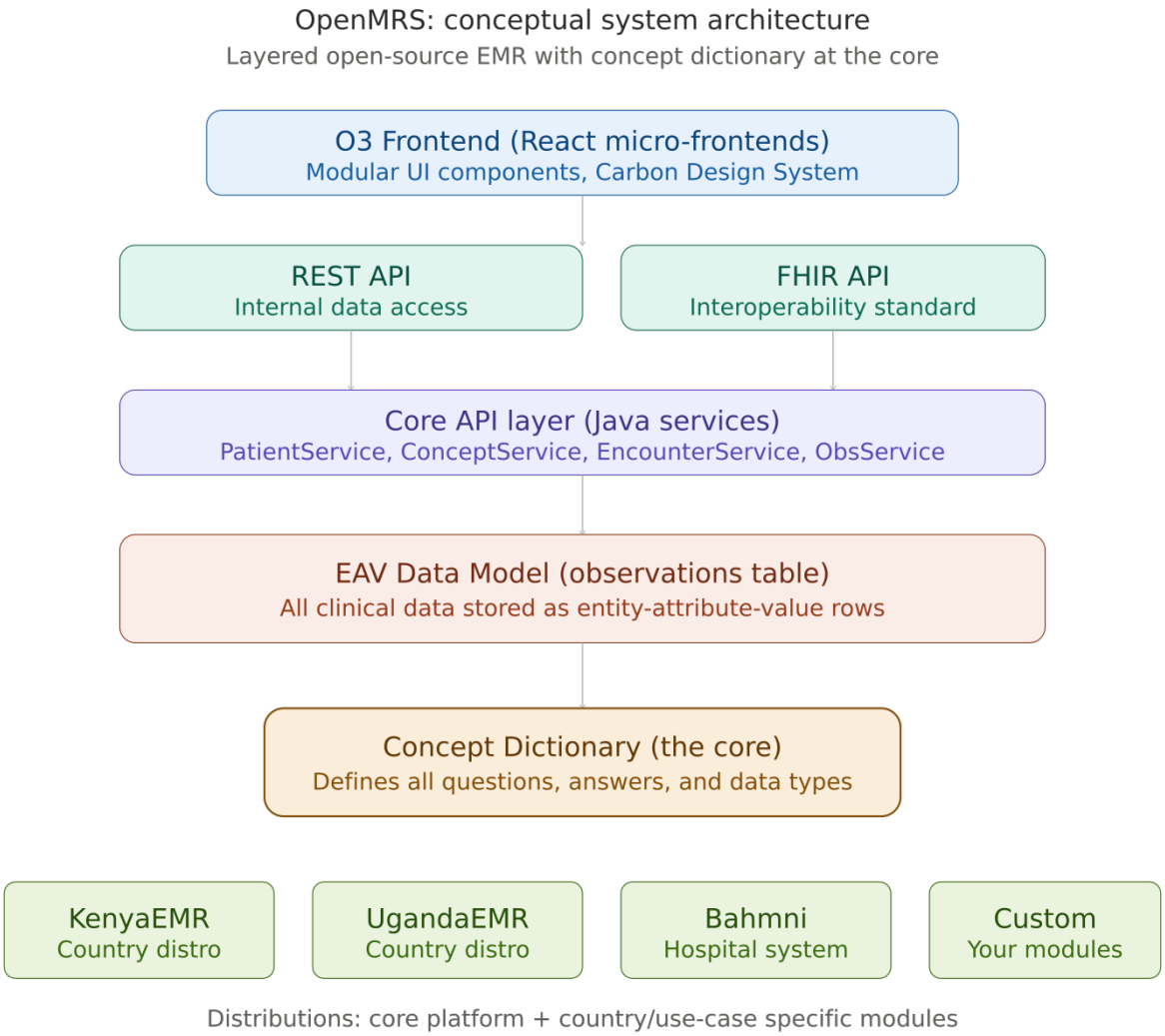
Existing EMR systems (Epic, Cerner) cost millions to deploy, require dedicated IT teams, and depend on reliable infrastructure. OpenMRS was created to fill that gap: a free, open-source, modular EMR that can run on minimal infrastructure and supports the full range of clinical workflows.

> *Why this matters for an AI PM: Every AI-in-healthcare startup must answer "how does this connect to the patient record?" In high-income countries, the answer is proprietary (Epic, Cerner). In the 40+ countries running OpenMRS, the answer is an open system with a public data model you can study. Understanding OpenMRS means understanding healthcare data architecture.*

User Type	What They Need
Ministries of Health	National EMR standards and unified reporting
NGOs and aid organizations	Affordable care tools for low-resource settings
Hospitals/clinics in LMICs	Patient records without vendor lock-in
Global health researchers	Structured, cross-site clinical data
Developers	Open platform to build healthcare apps on

Architecture Analysis

OpenMRS has three architectural components that any AI PM in healthcare should understand.



Conceptual architecture based on OpenMRS documentation, O3 docs, and technical wiki

1. The Concept Dictionary

The heart of OpenMRS. A dictionary defining every data point: clinical findings, lab results, diagnoses, medications. Both questions and answers are encoded as concepts. The database schema never changes when new data types are needed. The concept dictionary makes cross-site analytics and ML substantially more feasible by improving semantic consistency.

2. Modular Architecture

Core platform with pluggable modules (OMODs backend, ESMs frontend). Country-specific distributions (KenyaEMR, UgandaEMR) add modules without forking core code. O3 uses React micro-frontends with REST and FHIR APIs.

3. Entity-Attribute-Value Data Model

All clinical observations stored as rows in a single table, each linked to a concept. Adding new data types requires zero schema changes. Querying requires understanding concept relationships, not just table names. A paradigm shift for AI engineers used to relational databases.

Component	What It Does	AI Relevance
Concept Dictionary	Defines all clinical data points	Semantic consistency for ML
Modular Architecture	Pluggable features	AI modules added without disruption
EAV Data Model	Flexible clinical storage	Specialized ETL needed
FHIR APIs (O3)	Standardized access	AI apps integrate via open standard

Product Decision Analysis

Decision 1: Open Source Instead of Proprietary

What this enables: Any organization can deploy without procurement or vendor contracts. Developers worldwide contribute. Country distributions built without forking core code.

What this costs: No commercial revenue. Depends on grants, donations, and volunteer labor. Feature development pace constrained by funding, not market demand.

> *My analysis: For the global health context, open source was the only viable path. The trade-off is slower evolution and dependence on sustained donor commitment.*

Decision 2: Concept Dictionary over Fixed Schema

What this enables: New diseases, lab tests, and protocols added by updating the dictionary, not the schema. Essential for HIV treatment where guidelines changed frequently. Enables cross-site research through shared vocabulary.

What this costs: Complex queries. Reporting requires understanding concept relationships. Performance can degrade without careful indexing.

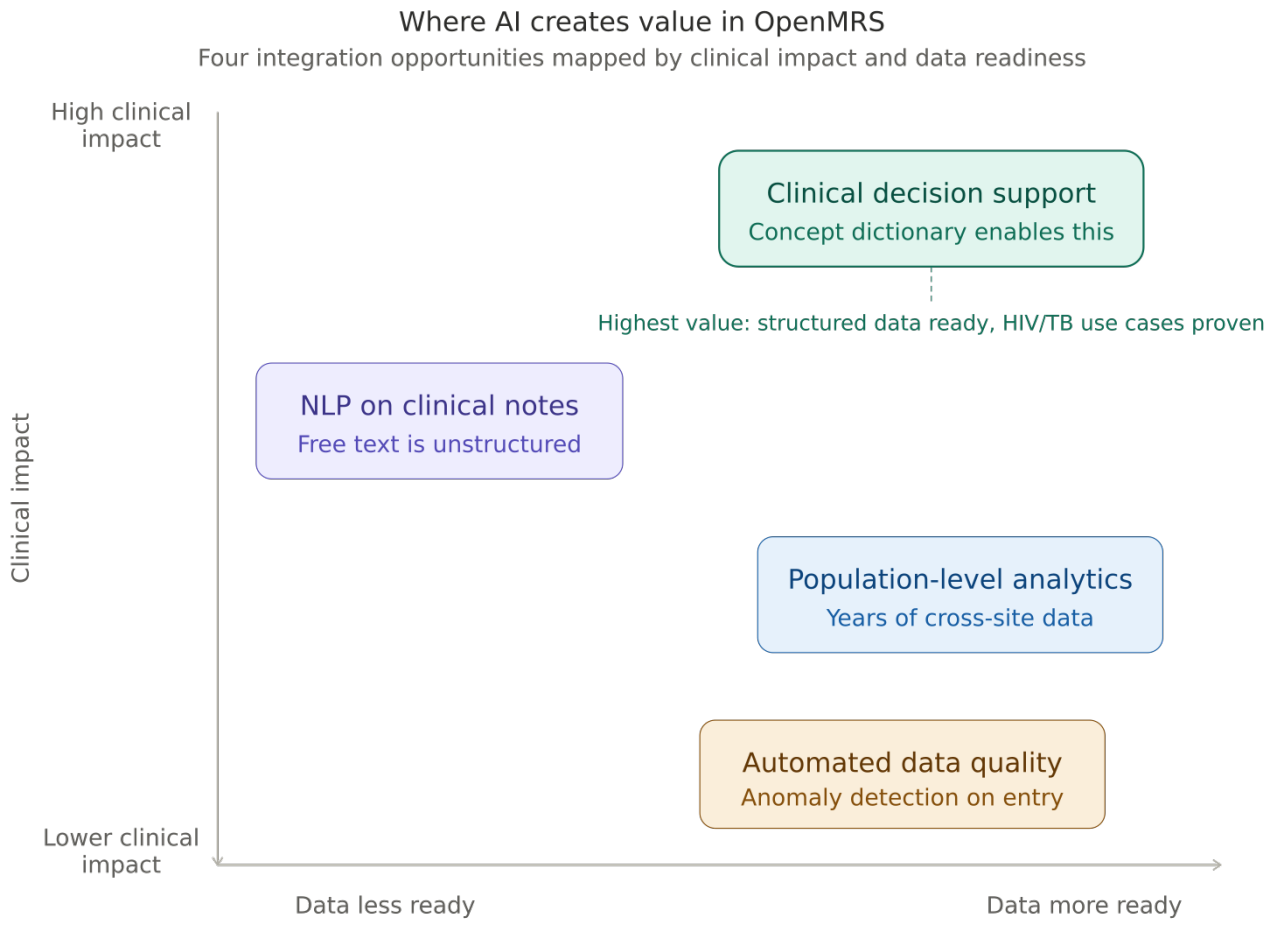
> *My analysis: A prescient decision. The concept dictionary makes OpenMRS data substantially more usable for research and analytics than unstructured clinical records.*

Decision 3: FHIR Compliance in O3

OpenMRS 3 introduced a FHIR module as a translation layer. The platform is described as “FHIR-friendly and FHIR-compliant” in its documentation. Any FHIR-speaking application can integrate with OpenMRS data.

> *My analysis: FHIR compliance transforms OpenMRS from “the EMR for developing countries” into an open-source, standards-compliant EMR that can participate in broader health data ecosystems. In principle (though integration always involves additional work), AI tools built for OpenMRS FHIR endpoints could be adapted for commercial EMR environments.*

Where AI Creates Value in OpenMRS



Assessment based on OpenMRS data architecture and published community discussions

1. Clinical Decision Support

Years of structured encounter data enable models that surface treatment recommendations based on similar past cases. For HIV, this means predicting therapy default risk. The concept dictionary makes this feasible because treatment data is semantically consistent.

2. NLP on Unstructured Clinical Notes

Clinicians write free-text notes containing insights invisible to the structured layer. NLP that maps extracted information to dictionary concepts would expand the data available for care coordination and research.

3. Predictive Analytics on Population Data

Population-level data on HIV, TB, and maternal health accumulated over years. In principle, predictive models could support forecasting of disease trends, identification of at-risk populations, and optimization of facility-level resource allocation. Realizing these outcomes in practice would require significant data cleaning, validation, and contextual adaptation.

4. Automated Data Quality

Missing fields, inconsistent coding, and entry errors are common. An AI layer detecting anomalies, suggesting corrections, and scoring quality in real time would improve both care and research reliability.

Key Metrics

North Star: Patient records with complete, structured data. The foundation for both care quality and AI readiness.

Metric	Why It Matters
Concept dictionary coverage	Coded vs free-text observations. Higher = more ML-ready data.
Data completeness per facility	Missing data limits care quality and research.
FHIR API adoption rate	Interoperability readiness across implementations.
Module ecosystem growth	New modules per year. Reflects community health.
Deployment time	Speed of new implementations. Reflects tooling maturity.

Sources

1. OpenMRS: Product page (features, deployment, estimated value): openmrs.org/about
2. Regenstrief Institute: OpenMRS founding history (PMC): ncbi.nlm.nih.gov
3. OpenMRS: O3 architecture documentation: o3-docs.openmrs.org
4. OpenMRS: Technical overview (data model, API, modules): wiki.openmrs.org
5. OpenMRS: O3 explained and investments: openmrs.org/o3
6. OpenMRS Talk: AI integrations (community discussion): talk.openmrs.org
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8. OpenMRS: Concept dictionary documentation: wiki.openmrs.org